Traffic Prediction System 🚦

A Machine Learning Approach to Intelligent Traffic Classification

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Group 4

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Course

DATA SCIENCE BEGINNERS

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Why Traffic Prediction Matters

Traffic congestion costs billions annually — in lost time, wasted fuel, and reduced productivity. Yet most drivers still navigate without real-time intelligence.

Travel Time

Commuters lose an average of 54 hours per year sitting in congestion.

Fuel Consumption

Stop-and-go traffic dramatically increases emissions and fuel waste.

Productivity

Delayed workers, missed deliveries, and economic ripple effects compound daily.

The Problem

Traffic behavior is inherently complex — shaped by weather, time of day, and human patterns. Current tools fall short.

Unpredictable Flow

Traffic surges without warning, making route planning unreliable.

Journey Planning Gaps

Drivers lack data-driven insight to make informed departure decisions.

Need for Smart Systems

A predictive, ML-powered solution can fill this critical intelligence gap.

Project Objectives

Build a Prediction Model

Train a classification model to distinguish High vs. Low traffic conditions.

Leverage Weather & Time

Incorporate temperature, rain, snow, hour, day, and month as input features.

Deploy an Interactive App

Build a Streamlit application for real-time user input and live predictions.

Dataset Features

Eight input features — drawn from weather and temporal data — feed the model, producing a binary **High / Low** traffic classification.

Temperature

Rain (mm)

Snow (mm)

Cloud Coverage

Hour

Day

Month

Weather Type

Target Variable: Binary classification — **High Traffic** or **Low Traffic**

Data Preprocessing Pipeline

Data Cleaning

Remove nulls and outliers

Encoding

Label-encode weather categories

Feature Scaling

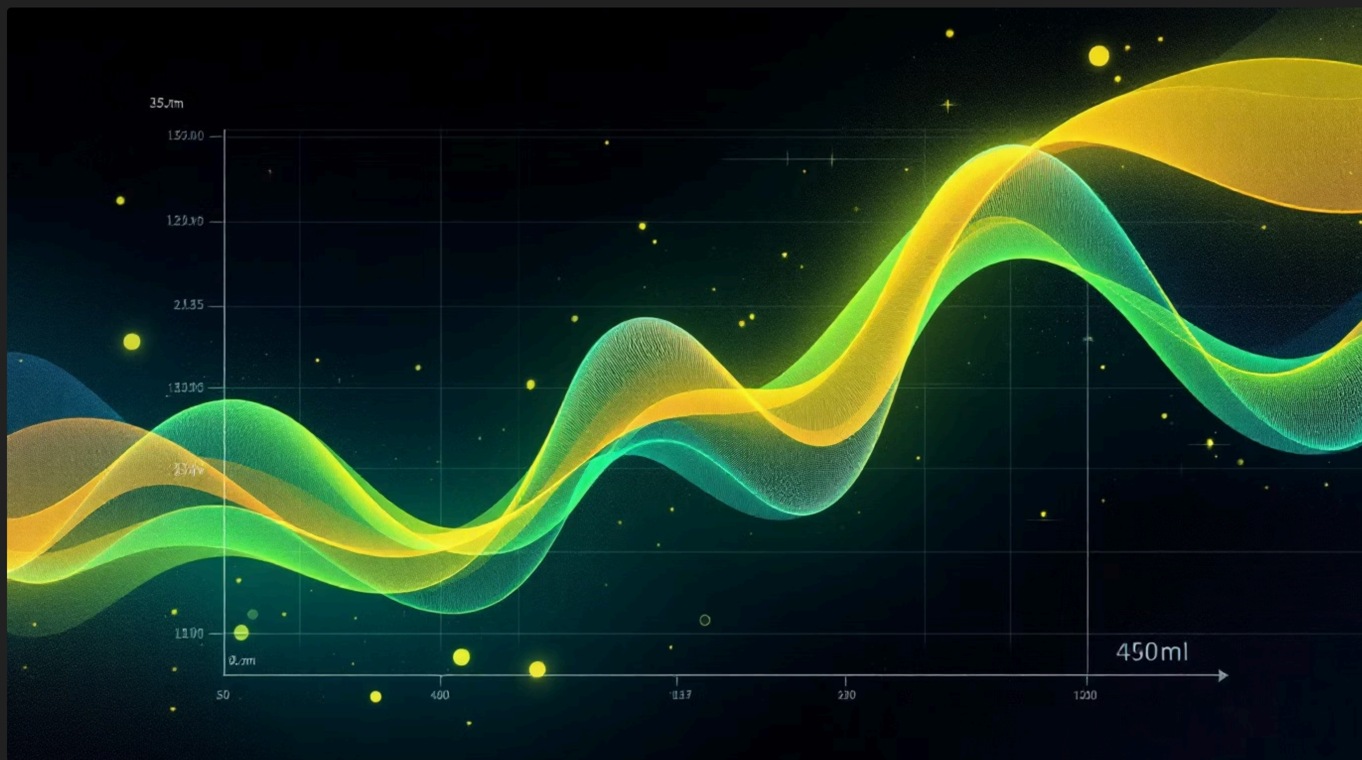
Apply StandardScaler normalization

Temp Conversion

Convert Celsius to Kelvin

Each step ensures the model receives clean, consistent, and numerically compatible inputs — critical for logistic regression to converge accurately.

Model: Logistic Regression



Logistic Regression was selected for its interpretability and efficiency on binary classification tasks.

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Estimates Probability

Outputs a probability score between 0 and 1 using the sigmoid function.

2

Classifies Traffic

Applies a threshold to classify the result as **High** or **Low** traffic.

3

Lightweight & Fast

Ideal for real-time prediction in a web application context.

Model Performance

66%

Classification Accuracy

Achieved on held-out test data across High and Low traffic classes.

2

Output Classes

Binary prediction — High Traffic vs. Low Traffic.

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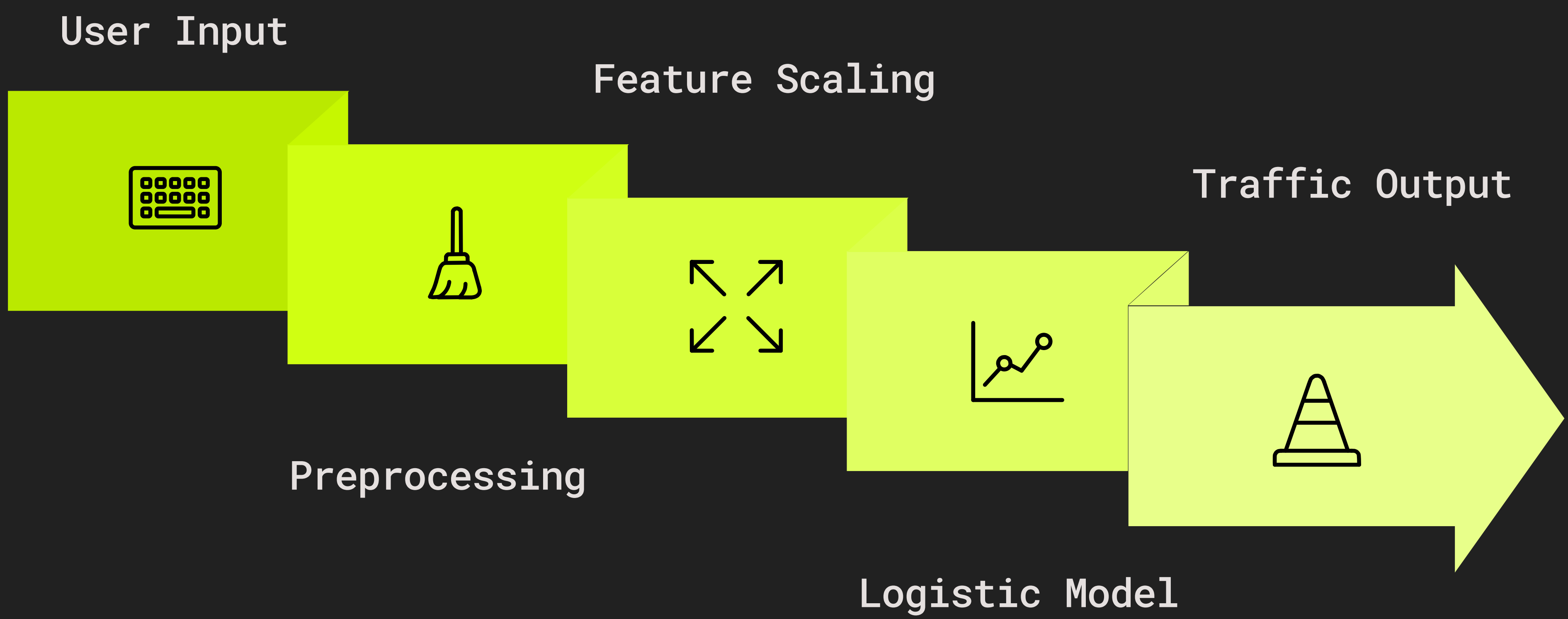
Input Features

Weather and temporal variables combined into a single feature vector.

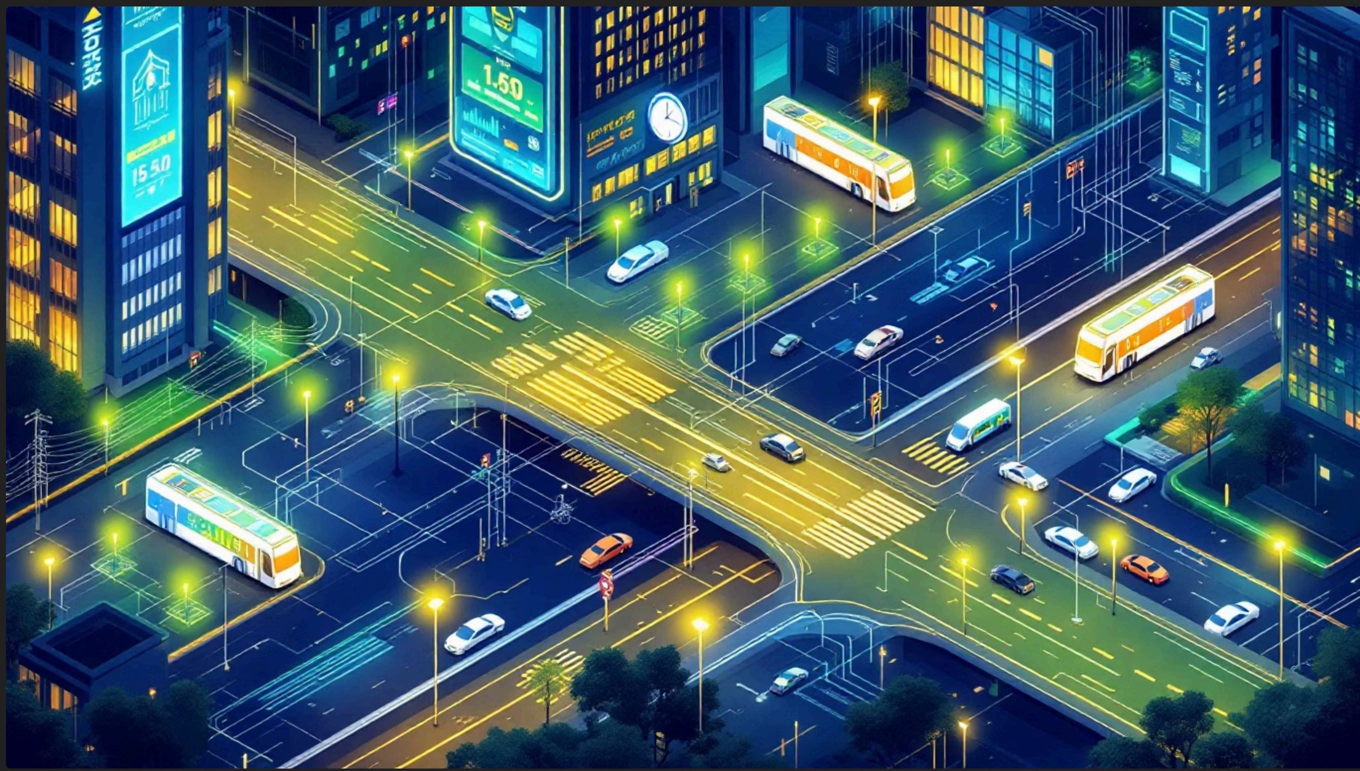
> Accuracy reflects baseline performance. Future iterations can incorporate ensemble methods or additional features for improvement.

System Flow



From raw user input to a classified prediction — the entire pipeline runs end-to-end in real time inside the Streamlit application.

Conclusion & Key Insights



The Traffic Prediction System demonstrates that even a lightweight ML model can extract meaningful patterns from weather and temporal data.

Rush Hour Drives Congestion

Hour and day features are the strongest predictors of high traffic.

Weather Compounds the Effect

Rain and snow measurably increase congestion probability.

A Foundation for Growth

The modular pipeline supports future upgrades — richer data, ensemble models, and deeper analytics.